Using Simulation to Find the Best Traffic-Light Timing Strategy

Course: DSCI 3411 Fundamentals of Simulations

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Contribution Statement	AI Use Statement
Both team members contributed equally to the project (50% / 50%). Youssef Sarofeem worked on simulation assumptions, R-model implementation, output generation, and numerical checks. Youssef Kiwan worked on model interpretation, report organization, validation discussion, and final consistency review. Both members reviewed the final report and source file.	AI assistance was used to help search for and refine a suitable project idea, connect the traffic-light queueing problem to the simulation topics covered in class, and organize the written report. AI was also used to help with some parts of the R implementation because the adaptive queueing logic, bootstrap summaries, and consistency checks were more difficult to implement using only the basic class examples. However, the team reviewed, edited, tested, and understood the final R code, including the simulation assumptions, random-variable generation, queue updates, strategy comparison, output tables, and plots. The final report, code, outputs, and conclusions represent our own work; AI was used only as support during parts of the development process. We reviewed, edited, understood, and take full responsibility for all submitted materials.

Abstract

This project studies a two-road traffic-light system as a stochastic queueing problem. Vehicles arrive randomly, vehicle types have different service headways, pedestrian requests occur randomly, and the signal alternates between green phases and lost time. Because the queues depend on arrivals, remaining phase time, vehicle mix, and pedestrian requests, the best timing rule is not obvious analytically. The project uses Monte Carlo simulation in R to compare five strategies: Equal 35/35, Short 30/20, Main Priority 50/25, Long Cycle 70/30, and Adaptive Pressure. The final recommendation is based on mean waiting time, high-percentile waiting time, fairness between roads, queue burden, and spillback risk.

1. Research Question

The research question is: which traffic-light timing strategy minimizes vehicle waiting time while keeping the side road reasonably fair and avoiding spillback? This matches the course idea of using simulation to answer a real-life problem and to compare competing methods when a simple closed-form mathematical proof is difficult.

2. Model Description

The intersection has a main road and a side road. Only one road receives green service at a time. During lost time, neither queue is served. The main road has higher arrival demand than the side road, and demand changes across normal, rush, and recovery periods. Cars, heavy vehicles, and buses require different service headways, so service time is random through the vehicle type.

The numerical inputs in this project are not meant to represent one exact real intersection. Instead, they are controlled simulation assumptions chosen to create a realistic traffic situation with unequal demand between the two roads. The main road is assumed to have higher traffic volume than the side road, and demand is allowed to rise during the middle part of the three-hour period to represent a rush period. This makes the experiment useful because the strategies are tested under a challenging but still understandable queueing setting.

Model item	Value	Purpose
Simulation length	10,800 sec	Three-hour study period
Warm-up	600 sec	Ignored for waiting-time summaries
Main-road rates	0.11, 0.19, 0.13 veh/sec	Normal, rush, recovery
Side-road rates	0.055, 0.095, 0.065 veh/sec	Lower demand than main road
Vehicle headways	2, 4, 6 sec	Car, heavy vehicle, bus
Lost time	4 sec plus pedestrian effect	Yellow/all-red clearance
Spillback threshold	55 vehicles	Risk measure for long queues

The exact arrival times, vehicle types, pedestrian effects, and waiting times are random outputs of the simulation. However, the arrival rates, service headways, lost time, and spillback threshold are fixed input assumptions. Keeping these inputs fixed across all strategies allows the comparison to be fair, because each timing strategy is tested under the same traffic environment. Therefore, differences in the results are mainly due to the signal-timing strategy rather than different traffic conditions.

3. Random Variables and Formulas

Interarrival times are exponential and generated by inverse transform: X = -log(1 - U) / lambda, where U is uniform on (0,1). Pedestrian requests use a Bernoulli rule: I = 1 if U < p and I = 0 otherwise. The empirical distribution function is F_n(x) = number of observations less than or equal to x divided by n. The time-average queue length is estimated as the area under the queue-length curve divided by the total simulation time.

4. Strategies and Simulation Algorithm

The simulation compares four fixed signal plans with one adaptive rule. The adaptive rule checks the two queues at the beginning of a cycle and changes the next green times according to queue pressure. This makes it a useful comparison against simpler fixed strategies.

Strategy	Rule	Reason for inclusion
Equal 35/35	35 sec main, 35 sec side	Baseline that treats roads equally
Short 30/20	30 sec main, 20 sec side	Shorter cycle reduces long reds
Main Priority 50/25	50 sec main, 25 sec side	Favors the higher-demand road
Long Cycle 70/30	70 sec main, 30 sec side	Fewer switch losses but longer reds
Adaptive Pressure	Queue-based green times	Responds to the current queues

The five strategies were selected to represent different timing ideas. Equal 35/35 represents a simple balanced rule, Short 30/20 represents a shorter cycle, Main Priority 50/25 gives more green time to the busier road, Long Cycle 70/30 reduces switching losses but may increase red-time waiting, and Adaptive Pressure changes the green times based on the observed queues. This range of strategies makes the comparison more meaningful because it includes both simple fixed plans and a queue-responsive plan.

Algorithm summary: set the random seed, generate arrivals and vehicle types, initialize the two queues, move the signal through green and lost-time phases, add arriving vehicles each second, serve vehicles during green according to headway, record waiting times and queue lengths, and repeat the experiment for each strategy.

5. Experimental Design and Validation

The final run uses 30 replications per strategy, giving 150 simulated three-hour traffic periods in the base experiment. The bootstrap confidence intervals use 200 resamples of the replication means. The sensitivity analysis uses 10 replications for each demand multiplier and strategy. This reduced design was chosen because it finishes in a reasonable time while still giving repeated-run evidence for comparing the strategies. The model was checked by verifying that empty queues produce zero waiting time, higher demand increases queues and waiting times, main-road priority lowers main-road delay while increasing side-road delay, and the queue traces show build-up during red phases and clearance during green phases.

Since simulation results change from one random run to another, the experiment uses repeated replications instead of relying on one run only. The confidence intervals and replication-level plots help show whether the differences between strategies are consistent. The sensitivity analysis also checks whether the conclusion changes when demand becomes lower or higher than the base case.

6. Main Results

The Adaptive Pressure strategy has the lowest mean waiting time and the best overall score. Main Priority also performs well on total waiting time, but it is less fair to side-road vehicles. Short 30/20 has the smallest fairness gap among the strong methods, but its average waiting time is higher than Adaptive Pressure. Equal 35/35 is not recommended because equal green time is not suitable when demand is unequal.

The score is used as a single summary measure to rank the strategies, but it should not be interpreted as the only result. It combines average waiting time, high-percentile waiting time, long-wait risk, fairness between the two roads, and spillback risk. Average waiting time receives the most attention because it measures overall efficiency, while the other terms penalize strategies that create unfairness or unusually long queues. The individual measures are still shown separately so that the recommendation does not depend only on one combined score.

Strategy	Mean wait (95% CI)	P90 / P95	Main / Side	Gap	AvgQ / MaxQ	Score
Adaptive Pressure	17.65 (17.40-17.89)	36.52 / 44.44	14.72 / 23.45	8.73	3.66 / 18.6	32.17
Short 30/20	21.00 (19.96-22.03)	44.88 / 59.61	22.16 / 18.61	4.48	4.33 / 24.2	37.68
Main Priority	19.31 (18.94-	44.65 / 53.80	14.61 / 28.71	14.10	4.01 / 18.6	37.76

Strategy	Mean wait (95% CI)	P90 / P95	Main / Side	Gap	AvgQ / MaxQ	Score
50/25	19.69)					
Long Cycle 70/30	22.50 (22.12-22.88)	57.19 / 71.02	13.94 / 39.45	25.51	4.72 / 19.1	47.64
Equal 35/35	45.50 (40.10-50.91)	126.76 / 160.55	59.83 / 16.69	43.14	9.44 / 43.5	100.90

The final ranking by score is Adaptive Pressure first, followed by Short 30/20 and Main Priority 50/25. In the base scenario, Adaptive Pressure has a mean wait of 17.65 seconds and a 90th-percentile wait of 36.52 seconds. Equal 35/35 has the worst mean wait (45.50 seconds) and the largest fairness gap, so it is clearly dominated under these assumptions.

7. Graphical Evidence

The figures support the numerical ranking. Adaptive Pressure keeps the waiting-time distribution lower and makes the queue process less extreme, while Equal 35/35 has both a high mean and a long right tail. The queue trace also connects to Monte Carlo integration because the time-average queue is the area under the queue-length curve divided by total time.

Figure 1. Average waiting time with confidence

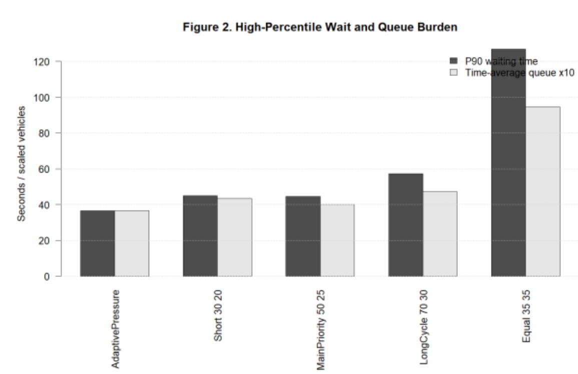

Figure 2. High-percentile waiting and queue

intervals.

burden.

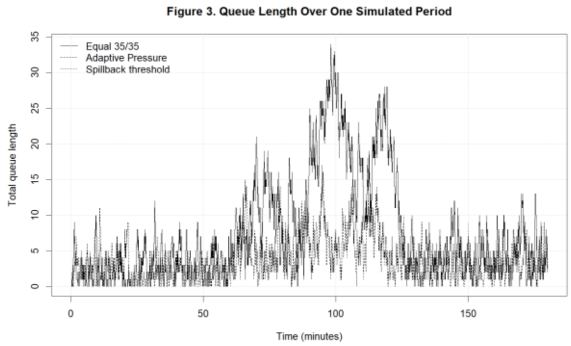

Figure 3. One simulated queue-length trace.

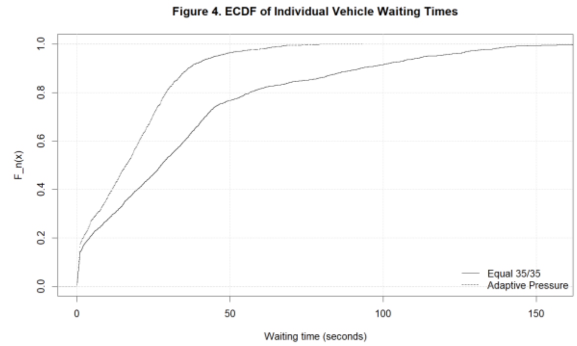

Figure 4. ECDF of individual waiting times.

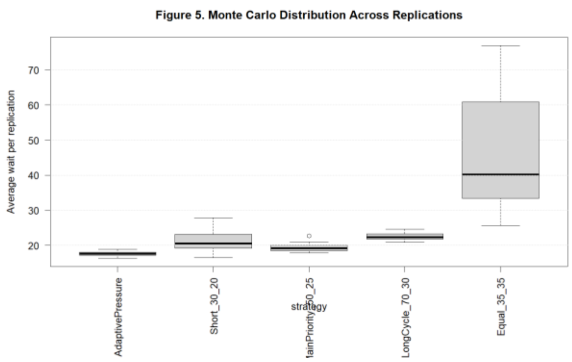

Figure 5. Replication-level average waiting times.

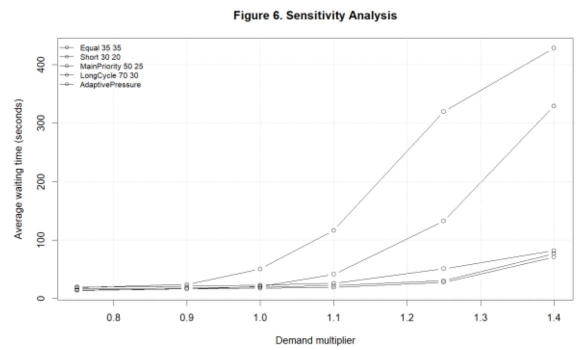

Figure 6. Sensitivity to demand multiplier.

8. Sensitivity Analysis

A strategy that works under one demand level may fail when traffic demand changes. The experiment therefore repeats the simulation under demand multipliers 0.75, 0.90, 1.00, 1.10, 1.25, and 1.40. The table below shows mean waiting time in seconds.

Demand	Adaptive	Short 30/20	Main Priority	Long Cycle	Equal 35/35
0.75	14.03	14.42	16.62	19.71	18.85
0.90	16.38	16.34	17.99	21.19	24.12
1.00	17.75	20.74	19.61	23.01	50.64
1.10	19.33	41.42	22.16	26.10	116.42
1.25	27.77	132.48	30.54	50.99	319.89
1.40	71.00	329.39	76.75	82.08	428.35

The sensitivity analysis confirms that Adaptive Pressure remains competitive as demand increases. At the highest multiplier, all strategies worsen, but Equal 35/35 and Short 30/20 deteriorate much more sharply than Adaptive Pressure, Main Priority, and Long Cycle. This supports the recommendation that a queue-responsive rule is more robust than a fixed equal split.

9. Limitations and Conclusion

The model is intentionally simplified for a simulation-course project. It uses two approaches rather than a full four-way intersection, assumes arrival rates instead of estimating them from field data, and does not include turning lanes, lane changes, accidents, blocking from downstream intersections, or emergency vehicles. The adaptive rule is also simple so that it stays close to the course material.

Overall, Adaptive Pressure is recommended because it has the best efficiency-reliability balance in the base scenario and remains strong under changing demand. The project shows how simulation can provide repeatable evidence when a real queueing system is too complex for one clean formula. By comparing competing strategies using Monte Carlo replications, ECDFs, bootstrap intervals, and queue-length integration, the project answers the practical question of which signal timing rule performs best under the stated assumptions.